

**SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)**  
**Department of Computer Science and Engineering**  
**ASSESSMENT TOOL 1 – CONCEPT MAPPING**

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|                         |                                                                                                                                                                                         |                      |                                         |
|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|-----------------------------------------|
| <b>Course Code:</b>     | CSA65                                                                                                                                                                                   | <b>Course Title:</b> | Generative AI and Large Language Models |
| <b>CO Assessed:</b>     | CO1 – Precision (BL1, BL2)                                                                                                                                                              | <b>Assessment:</b>   | Assessment Tool 1 – Concept Mapping     |
| <b>Weightage:</b>       | 20% of CIA                                                                                                                                                                              | <b>Total Marks:</b>  | 100 (2 Questions × 50 Marks)            |
| <b>CO1 Statement:</b>   | <i>Explain the fundamentals of Artificial Intelligence, Generative AI, Foundation Models, Transformer architecture, tokenization, embeddings, and Large Language Models. (BL1, BL2)</i> |                      |                                         |
| <b>Student Name:</b>    | Y Jayesh Ranjan                                                                                                                                                                         | <b>Reg. No.:</b>     | 192472201                               |
| <b>Section / Batch:</b> | CSA - AI                                                                                                                                                                                | <b>Date:</b>         | 25/07/2026                              |

**Instructions to Students**

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

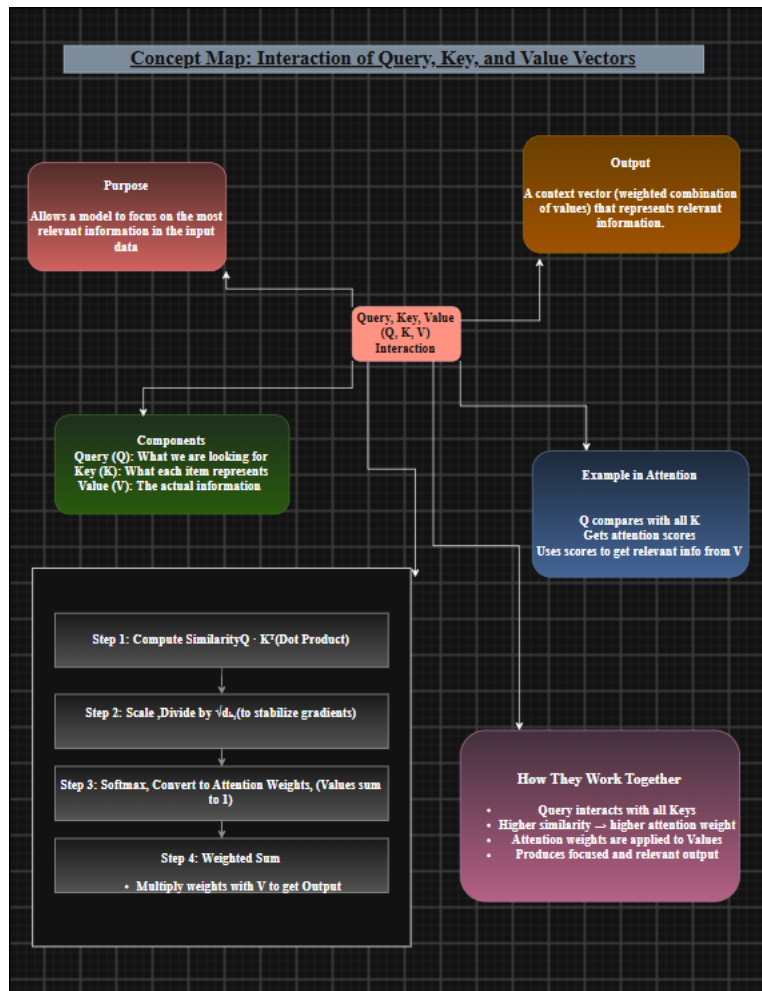
**Code of Conduct**

I Y. Jayesh Ranjan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

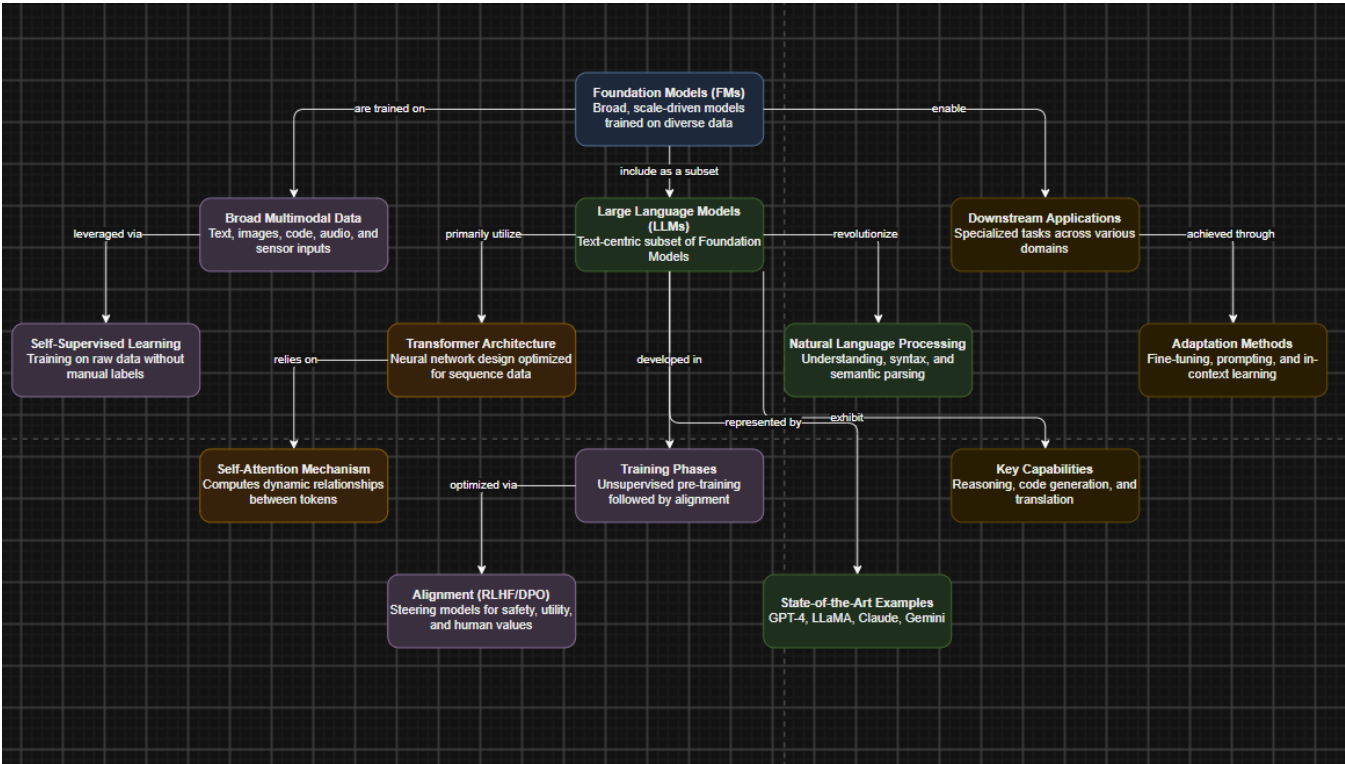
Signature of the Student: Y. Jayesh Ranjan

**SECTION A – CONCEPT MAPPING**

15. Develop a concept map explaining the interaction of Query, Key, and Value vectors.



26. Create a concept map showing the relationship between Foundation Models and Large Language Models.



**RUBRICS FOR CALCULATION**

| Criteria            | Weightage | Level 4 – Exemplary                                                                                   | Level 3 – Proficient                               | Level 2 – Developing                      | Level 1 – Beginning                     |
|---------------------|-----------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------|-------------------------------------------|-----------------------------------------|
| Technical / Concept | 30%       | All key concepts (AI, ML, DL, GenAI, LLMs) correctly identified with clear hierarchical relationships | Most concepts correct; minor gaps in relationships | Several concepts missing or misclassified | Major conceptual errors; map incomplete |

|                                      |     |                                                         |                                           |                                        |                                     |
|--------------------------------------|-----|---------------------------------------------------------|-------------------------------------------|----------------------------------------|-------------------------------------|
| <b>Application</b>                   | 25% | Real-world examples linked accurately to each concept   | Examples mostly relevant                  | Few or generic examples                | No real-world linkage shown         |
| <b>Quality</b>                       | 20% | Logical, well-organized structure with clear connectors | Mostly organized; minor structural issues | Some disorganization affecting clarity | Poorly structured, hard to follow   |
| <b>Presentation</b>                  | 15% | Visually clear, creative, easy to interpret             | Neat and readable                         | Cluttered or inconsistent formatting   | Untidy, difficult to interpret      |
| <b>Documentation / Communication</b> | 10% | Concise labels/notes that add clarity                   | Adequate labeling                         | Minimal or unclear labeling            | No supporting labels or explanation |

**Marks Evaluation:**

| Question No.        | Technical / Concept(30%) | Application(25%) | Quality (20%) | Presentation (15%) | Documentation / Communication (10%) | Total Marks (Each – 50) |
|---------------------|--------------------------|------------------|---------------|--------------------|-------------------------------------|-------------------------|
| Q1                  |                          |                  |               |                    |                                     |                         |
| Q2                  |                          |                  |               |                    |                                     |                         |
| Overall Total (100) |                          |                  |               |                    |                                     |                         |

| Faculty In-charge | Course Coordinator | Head of Department |
|-------------------|--------------------|--------------------|
|                   |                    |                    |
| Signature: _____  | Signature: _____   | Signature: _____   |
| Date: _____       | Date: _____        | Date: _____        |